

A 20mW 130-175GHz Phase Shifter with Meandered $\lambda/2$ TLINES in BiCMOS 55nm

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Abstract—This paper presents a compact and low-power D-Band four-quadrant active phase shifter using meandered $\lambda/2$ transmission-lines (TLINES) for phase inversion. TLINe impairments (loss and phase deviation from 180°) are analyzed and their effects on the output vector space are discussed. A calibration technique is applied such that the phase shifter RMS amplitude and phase errors are kept under 1.2 dB and 8° , respectively. Implemented in SiGe BiCMOS 55 nm, the proposed circuit covers 130 - 175 GHz bandwidth with -4.5 dB peak gain, OP_{1dB} greater than -3 dBm and 20 mW average power consumption.

Index Terms—Phase shifter, D-band, BiCMOS, Vector Interpolation, 5G

I. INTRODUCTION

THE continuous demand for high-throughput data-intensive applications, paired with advances in SiGe BiCMOS technologies, are pushing the evolution of wireless systems towards mmWave and sub-THz bands, where the wide available bandwidth enables wireless communications with a transfer rate of 100 Gb/s. In particular, D-Band (110 - 170 GHz) is gaining interest as candidate for high-capacity point-to-point links in the network infrastructure for 5G and beyond [1], showing favorable oxygen attenuation and practicable antenna size when compared to lower frequency bands. Due to the high path-loss, traditional radio architectures, with single high-gain directive antenna, set excessively tight specifications for the power amplifier and run into technology limitations. Beam-forming, with active phased array systems, overcomes the issue by providing high EIRP thanks to increased antenna gain and over-the-air power combining. In addition, the flexibility of beam steering relaxes the challenge of TX - RX alignment set by the narrow beam width.

In active phased arrays, the transmitted and received signal at each antenna element needs to be properly aligned in phase in order to achieve coherent power combining, and this is accomplished with programmable phase shifters, which can be implemented in both passive and active form. Passive phase shifters rely on programmable filters [2], reflective type structures [3] or switched delay paths [4] and offer high linearity with no power consumption, at the price of severe insertion loss, limited phase shift resolution and large area occupation. On the other hand, active phase shifters can easily cover the 360° circle with high phase resolution, limited loss and small footprint with a power consumption and linearity trade off.

Shown as block diagrams in Fig. 1, active phase shifters are based on the so-called vector interpolation principle, where quadrature-phase signals (I/Q) are weighted by variable-gain amplifiers (VGA) and properly combined to achieve the desired phase shift. To cover 4 quadrants, i.e. a 0° - 360° phase

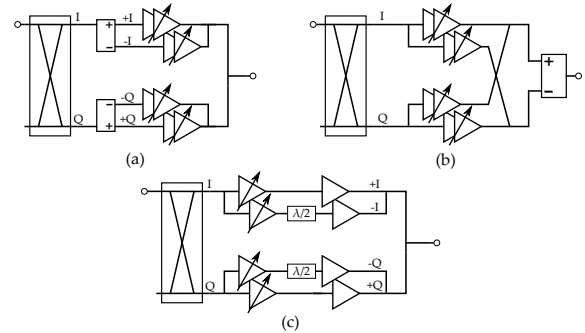


Fig. 1. Vector modulation phase shifters with balun (a) before and (b) after the VGAs. (c) Proposed architecture with $\lambda/2$ TLINES.

shift, two options are possible. In Fig. 1a, two baluns after the I/Q generation feed the VGAs with positive and negative signals ($\pm I, \pm Q$) [5]. Two of the four VGAs select the quadrant (i.e. the proper sign for I/Q) and the outputs are summed. In Fig. 1b, a single balun is used after the amplifiers and selection of the sum or difference between I and Q is made possible by cross-connecting the outputs of two VGAs [6], [7]. In both the approaches, baluns contribute to limiting the circuit bandwidth and are either bulky, if implemented with TLINES, or lossy if realized as lumped-element coupled inductors.

This work proposes a phase shifter with the architecture in Fig. 1c. VGAs are implemented as cascode amplifiers and, to get rid of the balun, phase inversion is achieved by introducing wideband $\lambda/2$ TLINES between the common-emitter (CE) and common-base (CB) devices of two VGAs. Realized in a 55 nm BiCMOS technology, the phase shifter covers the 130 - 175 GHz 3-dB bandwidth with -4.5 dB peak gain on the constant amplitude circle, and more than -3 dBm OP_{1dB} at center frequency. The power consumption ranges from 14 mW to 24 mW from 2V supply. Experimental results compare favorably against previous works.

II. CIRCUIT DESIGN

The schematic of the phase shifter is shown in Fig. 2. After the input pad, a 3 dB coupler splits the RF input signal into the in-phase (I) and quadrature (Q) components. At sub-THz frequency, the wavelength (λ) on silicon is short enough to enable practical realization of a distributed on-chip coupler, giving a remarkable advantage in terms of bandwidth and loss compared to the lumped-element counterpart. A coupled-line based hybrid coupler is chosen over the branch line version for its superior bandwidth and area advantage. The 3D view of the realized structure is shown in Fig. 3. The coupled lines are $245 \mu\text{m}$ long, corresponding to an equivalent electrical

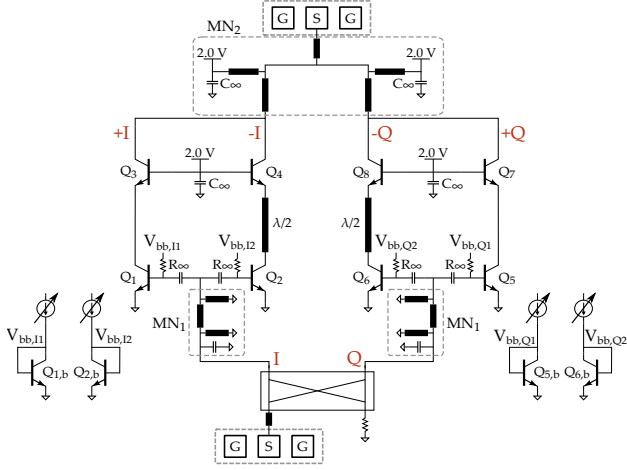


Fig. 2. Circuit schematic of the proposed vector modulator phase shifter.

length of $\lambda/4$ at 150 GHz. The 3 dB coupling factor between the I and Q output ports and $Z_0 = 40 \Omega$ are set by adjusting the width and spacing of the parallel lines, implemented in the topmost thick metal layer. The results of EM simulations are plotted in Fig. 3 bottom, showing $< 3^\circ$ phase error and < 0.7 dB amplitude imbalance over the 110 GHz - 200 GHz range. The insertion and return loss (not shown) are < 0.7 dB and > 15 dB respectively.

Looking again at Fig. 2, after the identical matching networks MN1, the I and Q signals feed the VGAs. Desirable features for the VGAs are a wide gain control range and high linearity, required to deliver high output power at the 1 dB gain compression point (OP_{1dB}) with good efficiency. Gilbert-cell analog multipliers are often used as VGAs in phase shifters [8], [9]. Their advantage is a superior dynamic range, at the expense of poor linearity at medium gain levels. In this design, simpler cascode amplifiers $Q_{1,3}$, $Q_{2,4}$, $Q_{5,7}$, $Q_{6,8}$ are used as VGAs. Similarly to [6], [7], [10], the magnitude of the signals $\pm I$, $\pm Q$ is controlled by simply regulating the bias current of the common-emitter transistors. This ensures the highest OP_{1dB} while at the same time minimizing the average power consumption, which is here proportional to the gain setting and not constant as in a Gilbert cell VGA. The common emitter (CE) and common base (CB) transistors in each cascode structure are equally sized with an area of $0.2 \times 8 \mu m^2$. The maximum bias current for each stage is 8 mA, set by current mirrors supplied by off-chip DACs. A matching network between the CE and CB transistors, often used to maximize gain by limiting the impedance mismatch, is not implemented. In fact, the parasitic base resistance of the CB transistors, r_b , leads to an inductive impedance at the emitter, $Z_{(e,CB)} \approx (1 + j\omega r_b c_\pi)/g_m$ (being g_m and c_π the transconductance and base-emitter capacitance). In the adopted technology, with similar sizing of the CE and CB, the positive reactance in $Z_{(e,CB)}$ well compensates the parasitic capacitance at the collector of the CE. At the same time, the feedback introduced by c_π reduces the output resistance of the CE, making it closer to the resistive component of

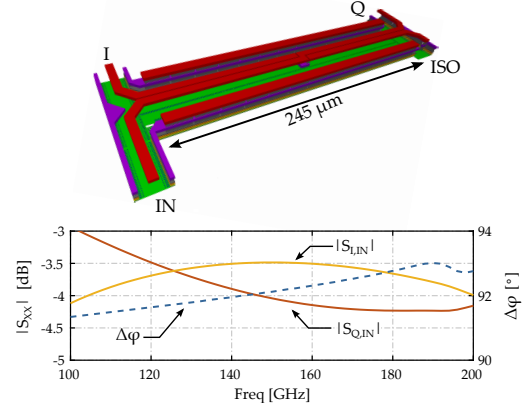


Fig. 3. 3D view and simulated performance of the hybrid coupler.

$Z_{(e,CB)}$. Thus, confirmed by simulations, an explicit matching network would bring only a marginal gain improvement with a penalty on the output compression point and thus it is not implemented.

The phase inversion on I and Q signals, required to cover the four quadrants, is performed by introducing simple transmission lines with an electrical length $\lambda/2$ at 150 GHz between the CE and CB transistors of two VGAs. In this way, the four vectors $\pm I$ and $\pm Q$ are available as current signals at the collectors of the four CB transistors. During operation, given the desired quadrant, only two VGAs are operated simultaneously while the others are kept off. The I and Q currents are summed with a TLINE-based combiner (MN2) which also sets the load impedance of the VGAs, selected as a compromise between gain and maximum linear output power.

A. Effect of $\lambda/2$ TLINEs amplitude and phase errors

The TLINEs used for phase inversion of the I and Q signals introduce attenuation and provide 180° phase shift only at the frequency where the electrical length is $\lambda/2$. Both the amplitude and phase errors on the inverted signals distort the magnitude and phase of the output vector, but the effect is compensated by calibration, which is anyway required to correct for the many other circuit non-idealities such as the imbalances of the quadrature hybrid, the non-linear gain control of the VGAs and the amplitude and phase distortion introduced by stray capacitances. The only effect of the $\lambda/2$ TLINEs errors which is not corrected by calibration is a reduction in the magnitude of the achievable constant gain circle, i.e. a penalty on the phase shifter gain.

If $+I$ and $+Q$ are the non-inverted vectors, available as currents at the collectors of Q_3, Q_7 in Fig. 2, the inverted signals at the collectors of Q_4, Q_8 can be written as $-I \cdot (1 - \varepsilon) \cdot e^{j\varphi_e}$, $-Q \cdot (1 - \varepsilon) \cdot e^{j\varphi_e}$ where ε and φ_e are, respectively, the amplitude and phase errors introduced by the $\lambda/2$ TLINEs. The effect of the two impairments is shown graphically in Fig. 4. Considering the TLINEs loss, i.e. the reduced amplitude of the inverted signals, the gray shadowed area in Fig. 4a represents

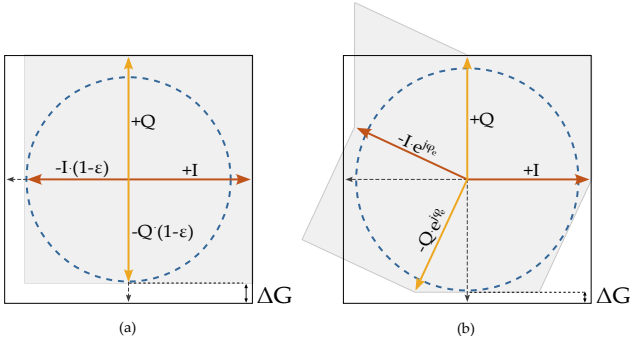


Fig. 4. Impact of the 180° amplitude (a) and phase (b) errors on the output polar characteristic.

the allowed vector interpolation space. The largest possible incircle centered at the origin is drawn with a dashed blue line. Its radius corresponds to the maximum achievable constant gain across the 0-360° phase shift. Compared to the ideal situation (represented by the external solid-line square) the gain is reduced by $\Delta G/G = \varepsilon$. The implemented $\lambda/2$ TLINES introduce ≈ 1 dB insertion loss which translates to the same penalty on the phase shifter available gain. Fig. 4b shows the effect of a phase rotation φ_e of the inverted signals. Also in this case, the radius of the largest incircle centered at the origin, within the shadowed region, is reduced. The gain penalty is $\Delta G/G = \sin(\varphi_e)$. The $\lambda/2$ TLINES introduce a constant time delay, therefore, a deviation from 180° phase shift is unavoidable over the band of interest. Within 130-175 GHz, the phase error is $\varphi_e = \pm 25^\circ$. Notably, despite the relatively large error, the penalty on the phase shifter gain is only 0.8 dB, i.e. less than the penalty introduced by TLINES loss.

III. EXPERIMENTAL RESULTS

The proposed phase shifter is realized in STM BiCMOS 55 nm technology. The chip photograph is reported in Fig. 5, with a core area of $400 \times 225 \mu\text{m}^2$ excluding GSG pads. To reduce area occupation, the $\lambda/2$ TLINES are meandered at the sides of the core. Small signal S-parameters are acquired by E8361C network analyzer through VDI WR 6.5 extenders. The complete space of the output vectors is measured via an automated routine by sweeping the VGAs gain over the entire range with off-chip 5-bit current DACs. Two VGAs are

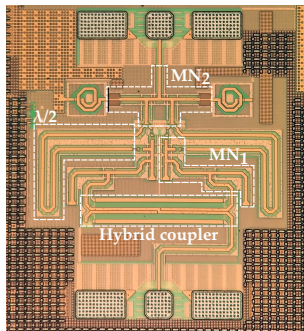


Fig. 5. Chip photograph.

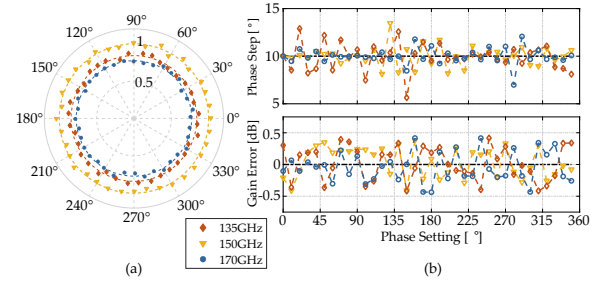


Fig. 6. Calibrated polar plot at 135-150-170 GHz, relative phase step and amplitude deviation from the ideal circle.

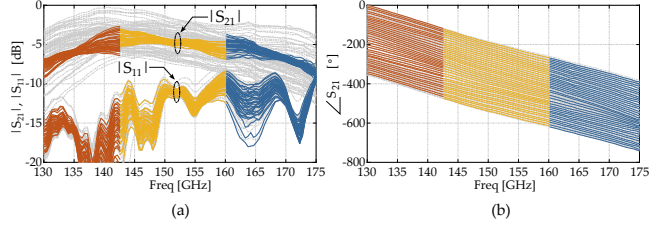


Fig. 7. Measured S_{21} and S_{11} (a) and phase response (b) of the interpolated states in the three calibration ranges.

simultaneously active while the other two are kept off. Each of the four possible VGAs combinations define the full 4-quadrant output map of the phase shifter which is therefore composed of $2^6 \times 2^6$ S-parameter measurements. Data is then processed in MATLAB and a calibration routine is applied at three different frequencies: 135 GHz, 150 GHz and 170 GHz. The VGAs control codes that best fit the constant gain circles with a phase step of 10° at the three frequencies are selected. The corresponding polar characteristic of the phase shifter is reported in Fig. 6a, where the circle radii are normalized to the magnitude at 150 GHz. The plots on the right in Fig. 6 report the point-to-point effective phase step, with an error lower than $\pm 4.5^\circ$, and the gain deviation from the ideal circle which remains lower than ± 0.5 dB.

Fig. 7a reports the magnitude of the transmission and input reflection coefficients (S_{21} , S_{11}) of the phase shifter across frequency, while Fig. 7b shows the transmission phase. The VGAs calibration settings selected at 135 GHz, 150 GHz and 170 GHz are considered valid in the three sub-bands: (1) 130-142 GHz, (2) 142-160 GHz, (3) 160-175 GHz. The corresponding curves are highlighted in different colors (red, yellow, and blue respectively). The average gain at the center frequency of 152 GHz is -4.5 dB and the gain decreases by less than 3-dB over the 130-175 GHz range. The calculated RMS gain and phase errors [11] across frequency in the three sub-bands are plotted in Fig. 8. The maximum RMS amplitude and phase errors are 1.2 dB and 8° , respectively.

The OP_{1dB} is measured at center frequency across the different phase settings in one quadrant and the plot is visible in Fig. 9. The compression point ranges from -3 dBm to -0.2 dBm. Depending on the desired output phase setting, the power consumption, from 2 V supply, varies from 14 mW to 24 mW with average of 20 mW. At $P_{out} = OP_{1dB}$ the power efficiency, η , is between 2.1 % and 3.5 %. Experimental results,

TABLE I
STATE OF THE ART COMPARISON TABLE

	This work	[6]	[7]	[12]	[13]	[8]	[5]
Technology	55 nm SiGe	130 nm SiGe	55 nm SiGe	130 nm SiGe	130 nm SiGe	130 nm SiGe	130 nm SiGe
$f_t/f_{\max}$ [GHz]	320 / 370	– / 450	320 / 370	200 / 220	– / 450	300 / 500	250 / 370
Frequency [GHz]	130 - 175	162 - 190	140 - 160	105 - 125	160 - 200	130 - 170	131 - 153
BW [%]	30	16	13	17	22	27	16
Gain [dB]	-4.5	-6.2	-4.5	4*	-9.5	4	-11
RMS phase err. [°]	8	8	7.5	9	15	-	-
RMS ampl. err. [dB]	1.2	1	1.4	2	0.9	-	-
OP _{1dB} [dBm]	-3	-27.2	-3.7	-10	-	-	-
IP _{1dB} [dBm]	2.5	-20	2	-8.5	-	-	-
Core area [mm ²]	0.09	0.07	0.05	-	0.075	-	0.14
P _{DC} /V _{CC} [mW/V]	[14 - 24]/2	[9.9 - 15.3]/1.8	50 [§] /3.3	44.6*/2.2	8.6 [§] /2.4	66/3.3	25/2.5
η at P1dB [%]	[2.1 - 3.5]	[0.01 - 0.02]	0.9 [§]	0.22	-	-	-

* Including buffer/s § Average across phase setting

summarized in Table I, compare favorably against previous works in the same band. Worth to be highlighted are the wide fractional bandwidth of 30 % and the high OP_{1dB} at low power consumption that leads to the highest reported efficiency.

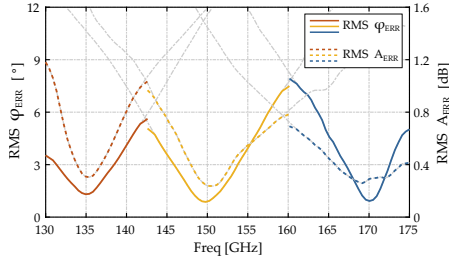


Fig. 8. RMS phase and amplitude errors across frequency.

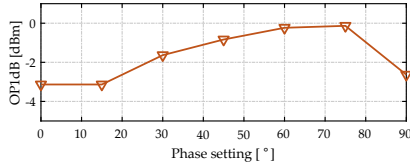


Fig. 9. Output compression point OP_{1dB} at 150 GHz in one quadrant.

IV. CONCLUSIONS

A vector interpolation phase shifter where baluns are replaced with $\lambda/2$ TLINEs phase inverters is presented. The impact of the TLINEs impairments are analyzed and considered. A compact D-Band active phase shifter is realized in BiCMOS technology proving a bandwidth, linearity and power efficiency enhancement with respect to similar solutions employing baluns. After calibration, the presented circuit provides constant-magnitude output circles with -4.5 dB peak gain over the 130-175 GHz frequency range. OP_{1dB} above -3 dBm is reached with an average power consumption of 20 mW from 2 V supply.

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